SHORT ANSWER TEXT

QUESTIONS		
Q.No	Question	Bloom's Levr
1	**Vehicle Management System** A transportation company maintains a database for managing different types of vehicles. Every vehicle has common information such as Vehicle_ID, Registration_Number, Manufacturer, Model, Year, and Purchase_Price. The company classifies vehicles into two specialized categories: Car and Truck. A Car has additional attributes such as Number_of_Doors, Fuel_Type, and Seating_Capacity, while a Truck has attributes such as Load_Capacity, Truck_Type, and Number_of_Axles. Both Car and Truck inherit the common attributes of the Vehicle entity through a specialization/generalization relationship. The specialization is disjoint, because a vehicle is classified as either a Car or a Truck, and it is total, because every vehicle in the system must belong to one of these two categories. Based on these requirements, construct an EER diagram showing the Vehicle supertype, Car and Truck subtypes, inherited attributes, specialization constraints, primary keys, and appropriate relationships. Tasks 1. Identify the supertype and subtypes. 2. Identify the common attributes of Vehicle. 3. Identify the specialized attributes of Car and Truck. 4. Specify the primary key of Vehicle and show its inheritance by the subtypes. 5. Represent the ISA specialization relationship. 6. Specify whether the specialization is disjoint or overlapping. 7. Specify whether the specialization is total or partial. 8. Construct the complete EER diagram using appropriate EER notation.	BL1 – Remember
2	**Bank Management System** A bank management system maintains information about customers, bank accounts, branches, employees, transactions, loans, and loan payments. Each customer is identified by a unique	BL2 – Understand

	Customer_ID and may hold one or more accounts such as Savings or Current Accounts. Each account has attributes such as Account_No, Account_Type, Balance, and Opening_Date and is maintained by a particular bank branch. A branch is identified by Branch_ID and stores Branch_Name, Location, and IFSC_Code. Each branch employs several employees, with details such as Employee_ID, Employee_Name, Designation, and Salary. Customers can perform multiple transactions such as deposits and withdrawals, where Transaction_ID, Transaction_Date, Transaction_Type, and Amount are recorded. A customer may also apply for loans, with details including Loan_ID, Loan_Type, Loan_Amount, Interest_Rate, and Loan_Status. Loan payments are recorded using Payment_ID, Payment_Date, and Payment_Amount. Based on these requirements, identify the entities, attributes, primary keys, and relationships with appropriate cardinalities, and construct a complete ER diagram for the Bank Management System. Tasks: 1. Identify the major entities. 2. List suitable attributes for each entity. 3. Specify the primary key of each entity. 4. Identify relationships among the entities. 5. Specify the cardinality of each relationship. 6. Construct the ER diagram using appropriate entities, attributes, relationships, and cardinalities.	
3	**Ecommerce Website** An E-Commerce Website maintains information about customers, products, orders, payments, and deliveries. Each Customer is identified by a unique Customer_ID and has attributes such as Name, Email, Phone, and Address. Customers can browse and purchase different Products, where each product has a unique Product_ID, Product_Name, Price, Stock_Quantity, and Description. Products are classified into different categories such as Electronics, Clothing, and Grocery, where each category may have its own specialized attributes. A customer can place multiple Orders, and each order is identified by Order_ID with attributes such as Order_Date, Order_Status, and Total_Amount. An order may contain multiple products, and the Quantity and Unit_Price of each ordered product must be recorded. Each order is associated with a Payment, containing Payment_ID, Payment_Date, Payment_Mode, and Payment_Status. The order is also associated with a Delivery, containing Delivery_ID, Delivery_Date,	BL2 – Understand

	Delivery_Status, and Delivery_Address. Based on these requirements, construct an EER diagram using specialization/generalization for product categories and show the appropriate entities, attributes, relationships, keys, and cardinalities. **Tasks** 1. Identify the major entities: Customer, Product, Category, Order, Payment, and Delivery. 2. Identify the appropriate attributes and primary keys. 3. Model the Order–Product relationship and include Quantity and Unit_Price as relationship attributes. 4. Apply specialization/generalization to classify Product into suitable subtypes. 5. Show the inherited attributes of the Product subtypes. 6. Specify whether the specialization is disjoint/overlapping and total/partial. 7. Represent the relationships between Customer–Order, Order–Payment, and Order–Delivery. 8. Specify suitable cardinalities for all relationships. 9. Construct a complete EER diagram using standard EER notation.	
4	**Hospital Management System – EER Diagram** A hospital management system maintains information about patients, doctors, nurses, departments, appointments, treatments, and medical records. Every hospital staff member has common attributes such as Staff_ID, Name, Address, Phone, and Salary. The hospital staff is specialized into Doctor, Nurse, and Administrative_Staff, where each subtype has its own specific attributes. A Doctor has attributes such as Specialization and License_Number, while a Nurse has attributes such as Qualification and Shift. The hospital also maintains Patient information including Patient_ID, Name, Date_of_Birth, Gender, and Contact_Number. A patient may have multiple medical records, but each Medical_Record is dependent on a particular patient for its identification and is therefore modeled as a weak entity with attributes such as Record_Date, Diagnosis, and Treatment_Details. Patients can consult doctors through Appointments, and doctors	BL2 – Understand

	can provide treatments to patients. The hospital is organized into several departments, with doctors assigned to specific departments. Based on these requirements, design an EER diagram incorporating specialization/generalization, inheritance, weak entities, relationships, and appropriate cardinalities. **Tasks** 1. Identify the major entities such as Staff, Patient, Department, Appointment, and Medical_Record. 2. Apply specialization/generalization to represent: o Staff → Doctor o Staff → Nurse o Staff → Administrative_Staff 3. Show the inherited attributes of Doctor, Nurse, and Administrative_Staff from Staff. 4. Identify the specific attributes of each subtype. 5. Represent Medical_Record as a weak entity dependent on Patient. 6. Identify the partial key of the weak entity. 7. Represent the identifying relationship between Patient and Medical_Record. 8. Model the relationships among Doctor, Patient, Appointment, and Department. 9. Specify appropriate cardinalities and participation constraints. 10. Construct a complete EER diagram using specialization/generalization and weak-entity notation.	
5	**Database Administrator (DBA)** A Database Administrator (DBA) is responsible for managing, maintaining, securing, and ensuring the reliable operation of an organization's database systems. In a large organization, databases contain critical information such as customer records, employee details, financial transactions, inventory, and business operations. The DBA ensures that this data is stored efficiently, remains	BL1 – Remember

ANSWER TEXT

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Vehicle Management System:

(i) Supertype and Subtypes:

- Supertype: vehicles
- Subtypes : Car, Truck

90/100

Airplane
Second hand

(ii) Common Attribute of Vehicles:

Vehicle_ID, Reg_No, Manufacturer, Model, Year, Purchase_Price.

(iii) Specialization Attributes:

- Car : Number_of_Doors, Fuel_Type,
- Truck: Load_capacity, Truck_Type, No_of_Axles

(iv) Primary Key:

Vehicle_ID

(v) ISA Specialization:

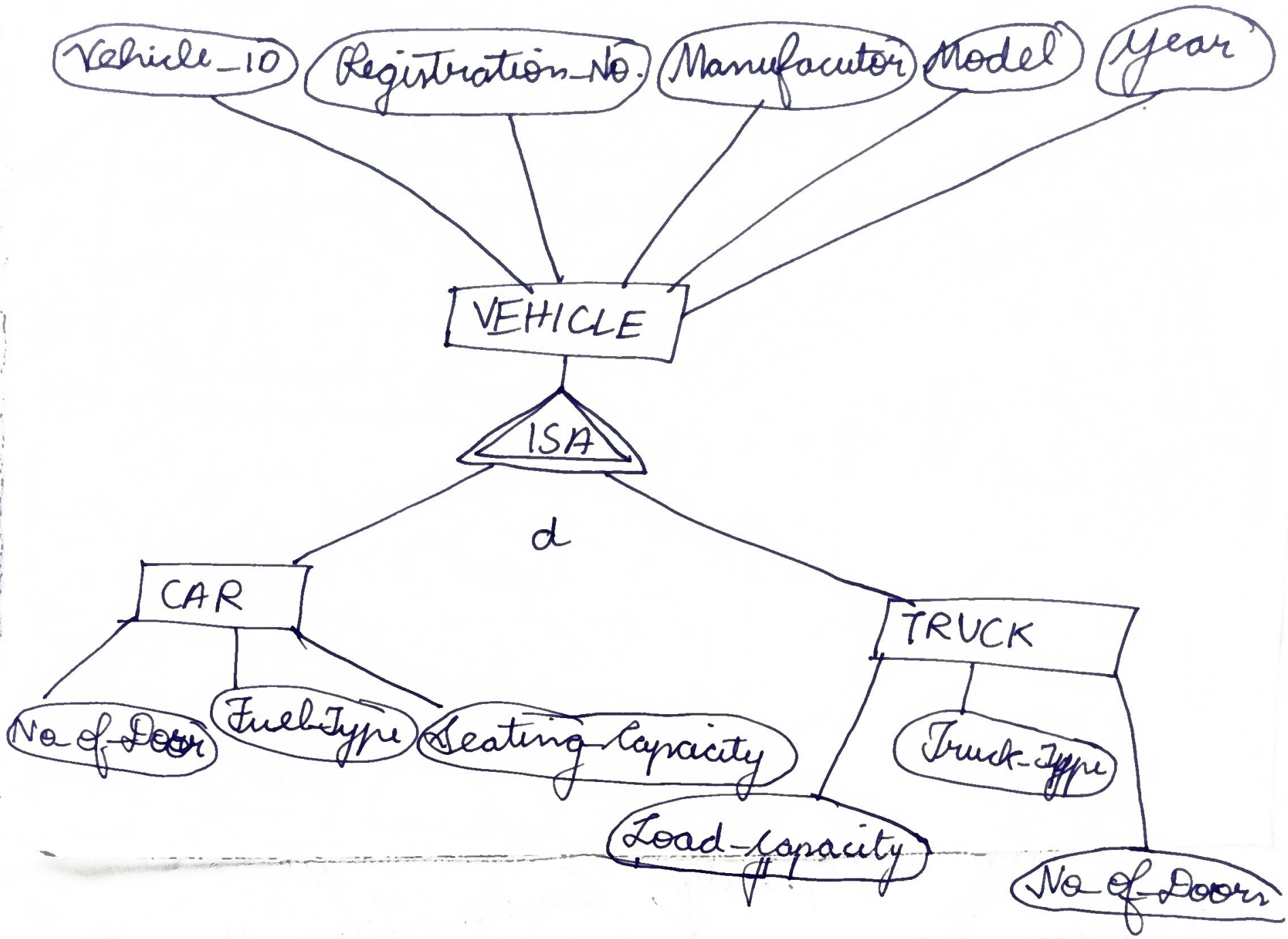
Vehical → Car or Vehicle → Truck

(vi) Disjoint: a vehicle is either a car or a Truck, never both

(vii) Total:

Every vehicle must belong to one subtypes

Vehicle Management Systems – EER Diagram



2: Bank Management System is Marked by ER Diagram

(i) Major entities

Customer, Account, Branch, employee, Transaction loan — Payment

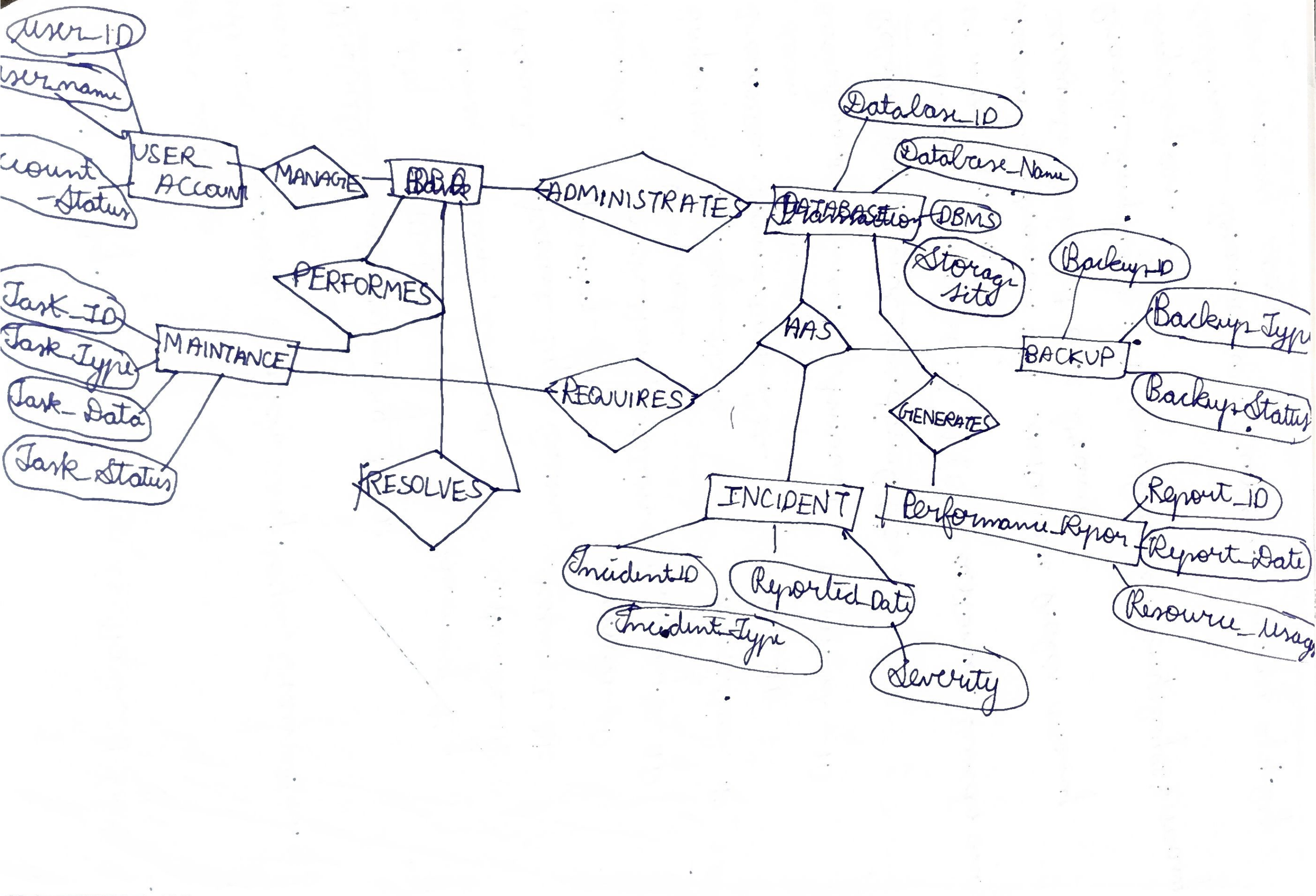
(ii) Attributes and primary keys:

Entity	Attributes	Primary Key
Customer	Name, contact	Customer_ID
Account	Account_Type, Balance, Opening_Date	Account_No
Branch	Branch_Name, Location, IFSC code	Branch_ID
Employee	Employee_Name	Employee_ID
Transaction	Transaction_Date	Transaction_ID
Loan	Loan_Type	Loan_ID
Loan_Payment	Payment_Amount	Payment_ID

(iii) Relationships and cardinalities:

* Customer — Account: 1 M (One customer holds one or more accounts)
* Account — Branch: M customer holds many accounts held at one branch
* Branch-Employee: M (One branch employee's many employees.
* Customer — Transaction: (Customer may apply for many loans)

Database Management System - EER



E-commerce Website - EER Diagram:

(i) Major Entities:

Customer, Product, category, Order, Payment, Delivery.

(ii) Attributes & Primary Keys:

Entity	Attributes
Customer	Name, Email, Phone, Address
Product	Product_name, Price, Stock_Quantity, Description
Category	Category_Name, specilized attributes.
Order	Order_Data, Order_Status, Total Amount
Payment	Payment_Date, Payment_mode, Payment_Status
Delivery	Delivery Date

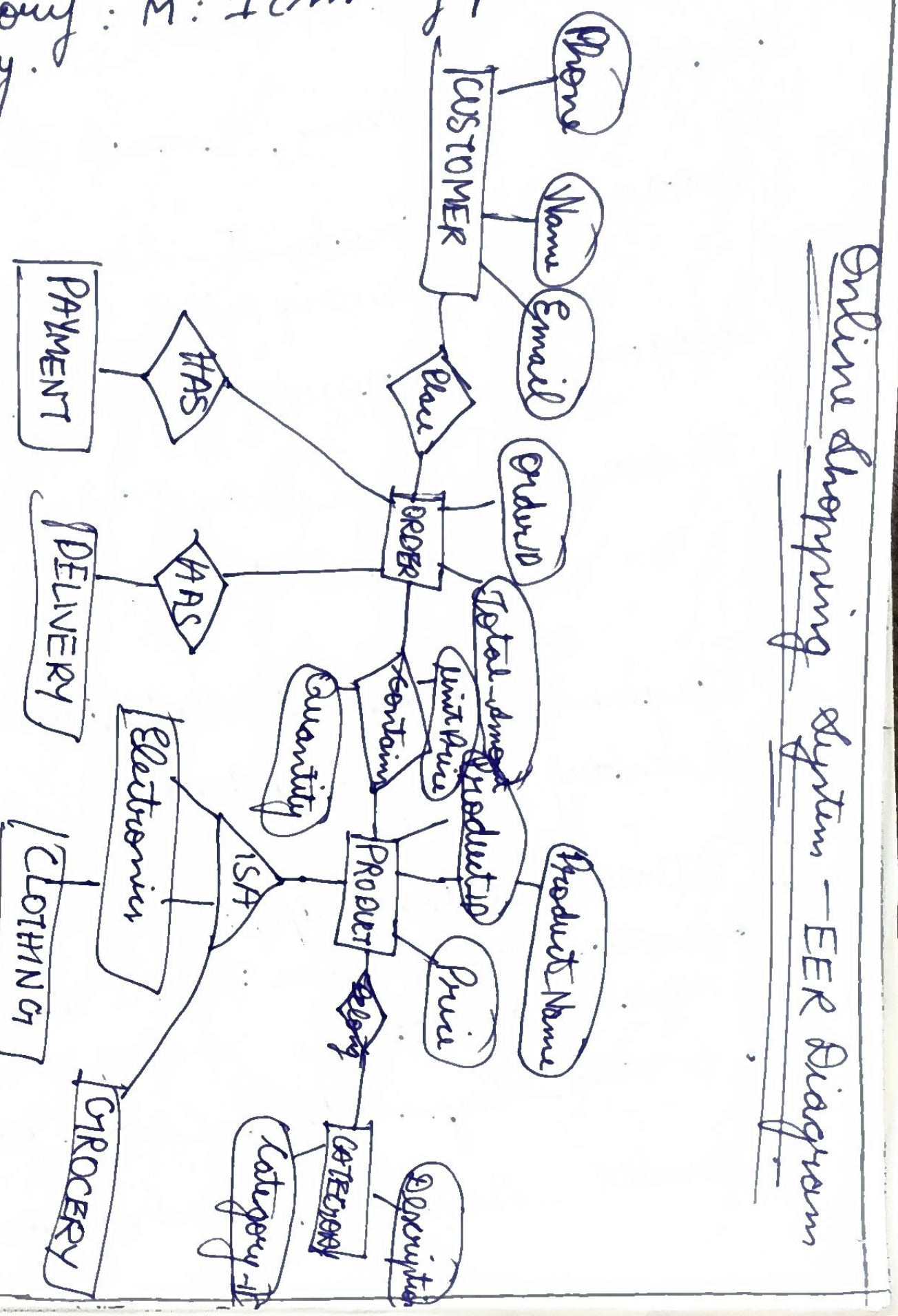
(iii) Other Relationships:

- Customer — Order : (One more customer place many orderes)
- Order_Payment : 1 (each other has one payment)
- Order_Delivery : 1 (each order has one Delivery)

(iv) Cardinalities Summary:

- Customer → Order : 1 : M
- Order → Product (contains) : M : N with Quantity & Unit_Price.
- Order → Payment : 1 : 1
- Order → Delivery : 1 : 1
- Product → Category : M : 1 (many products in one category.

Online Shopping System — EER Diagram



...agement System — EER Diagram:

(i) Major entities: Staff, Doctor, Nurse, Administrative Staff, Patient, Department, Appointment, Medical_Record

(ii) Specialization:
- Staff (Supertype) → Doctor, Nurse, Administrative_Staff
- ISA connector marked disjoint and total (every staff member fits a role)

(iii) Inherited attributes:
Staff – ID, Name, Address, Phone, Salary inherited by all three subtypes.

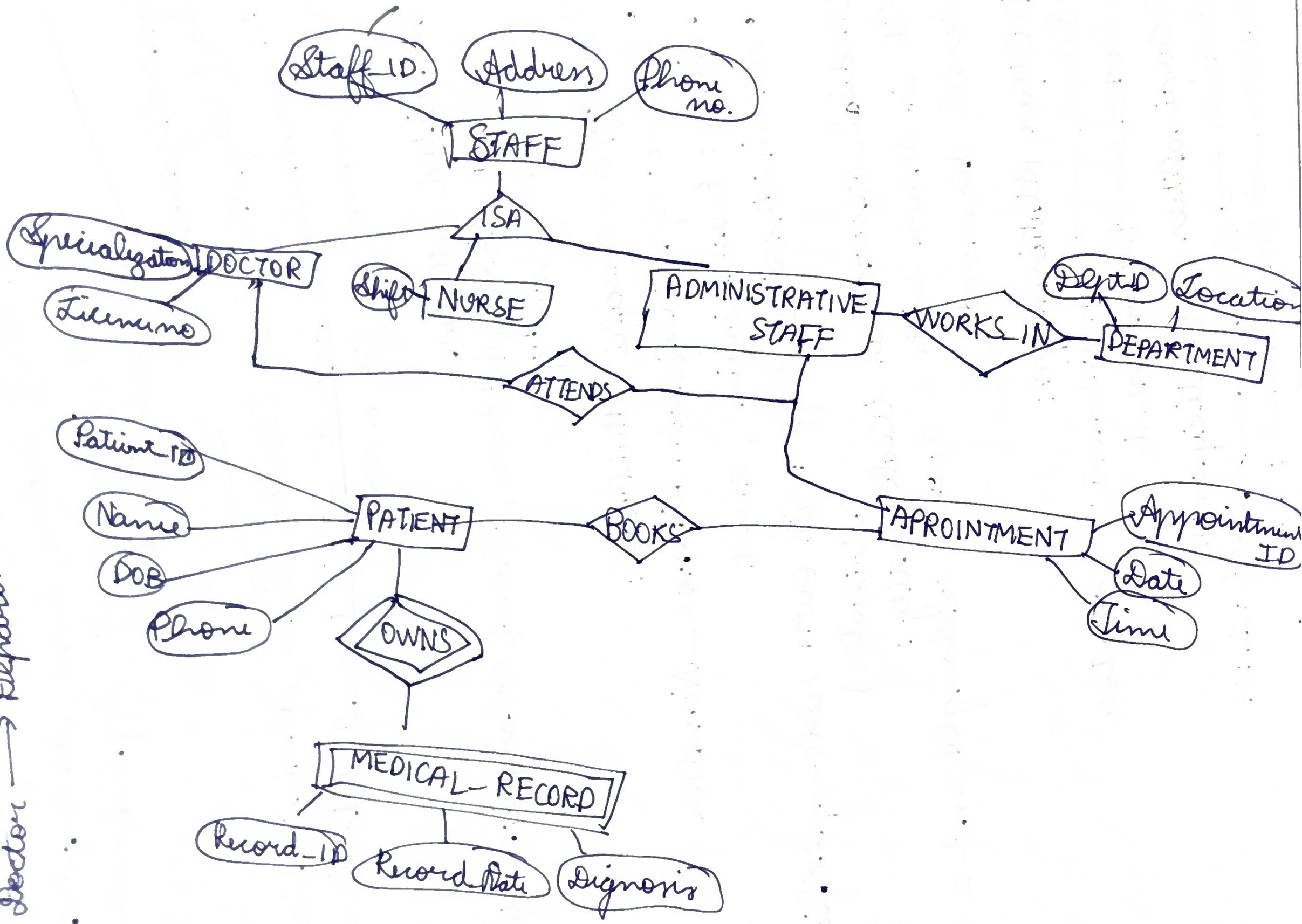
(iv) Subtype – Specific attributes:
- Doctor: Specialization, License_Number
- Nurse: Qualification, Shift
- Administrative_Staff: (its own role-specific attributes)

(v) Medical_Record as Weak entity:
It is drawn with a double rectangle and a double diamond.

(vi) Other Relationships:
- Patient → Appointment → Doctor
- Doctor → Department

(vii) Cardinalities on Participants:
- Patients → Medical_Record : 1 : M
- Doctors → Appointment : 1 : M

Hospital Management - EER Diagram



- Patient → Appointment: 1
- Doctor → Department: M : 1

Administrator (DBA)

(i) Role as DBA:

A Database Administrator manager, maintains, secures and ensures the reliable operation of an organization : database systems. They are responsible for keeping data.

(ii) Database design & implementation:

The DBA designs the database structure, installs and configures the DBMS, and implements it to meet organizational needs.

(iii) Security & user access control:

The DBA sets up authentication, assign roles privileges and controls who can view or modify data.

(iv) Backup & recovery:

The DBA implements backup scheduler and recovery strategies to protect data, against hardward failure.

(v) Performance Monitoring & Optimization:

The DBA monitors for slow queries, excessive resource usage, storage limits, then tunes indexes queries and configuration to improve performance.

(vi) Data Integrity & consistency:

The DBA enforces constrains, validation rules, and normalization to keep data accurate, consisten

(vii) Storage & Maintanance:

The DBA manages storage allocation, database sizing and routine maintainance like defragmentation and log management.

Database management Systems — EER Diagram

